



**Laboratory Manual
for
SC1003
Introduction to Computational Thinking and
Programming**

**Practical Exercise #7:
Python Virtual Environment and Jupyter Notebook**

**COLLEGE OF COMPUTING AND DATA SCIENCE
NANYANG TECHNOLOGICAL UNIVERSITY**

Ex. #7 – Virtual Environment & Jupyter Notebook

Learning Objectives

Create virtual environment and interact with Jupyter Notebook.

Equipment and accessories required

PC/notebook with python compiler and command prompt (eg windows cmd) or terminal

It is recommended to work and install the virtual environment and jupyter notebook on your own PC.

1. Python Virtual Environment

A Python virtual environment is a self-contained directory that contains a Python installation for a particular version of Python, along with additional packages. It allows you to create isolated environments for different projects, ensuring that dependencies and packages do not interfere with each other. Here are some key points about Python virtual environments

In this exercise, you will learn how to create a virtual environment and install some packages i.e. Jupyter Notebook. You will then explore and use jupyter notebook to write some simple program and describe or document the program with Markdown language.

2. Create & Interact with a Python Virtual Environment

1. Open command prompt or cmd or terminal (mac/linux users) and go to desktop
2. Create a directory and go into the directory:

```
> mkdir lab7
```

```
> cd lab7
```

3. Create a new environment

```
> python -m venv .venv (windows user)
```

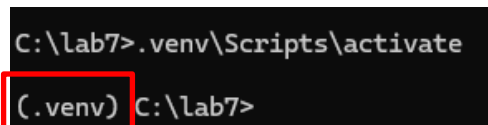
```
$ python3 -m venv .venv (mac/linux user)
```

4. Activate the created environment

```
> .venv\Scripts\activate (windows user)
```

```
$ source .venv/bin/activate (mac/linux user)
```

There should be a (.venv) on the left of your prompt looks like this.



```
C:\lab7>.venv\Scripts\activate  
(.venv) C:\lab7>
```

5. Check new environment packages

```
(.venv) > pip list
```

By default, the environment should be almost empty, and having only 1 package (pip)

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```
(.venv) C:\lab7>pip list
Package Version
-----
pip      24.2
(.venv) C:\lab7>
```

6. You have successfully created a new and fresh virtual environment.
7. To deactivate the virtual environment, execute the command `deactivate`

3. Installing & Starting a Jupyter Notebook

1. Reactivate the virtual environment that you have created just now.
2. Install the jupyter notebook package, the command below will not only install the jupyter notebook, but it will also install all other dependencies. This will take a while to complete.

```
(.venv) > pip install notebook
```

```
Installing collected packages: webencodings, wcwidth, pywin32, pure-eval, fastjsonschema, websocket-client, webcolors, urllib3, uri-template, types-python-dateutil, traitlets, tornado, tinycss2, soupsieve, sniffio, six, setuptools, send2trash, rpds-py, rfc3986-validator, pyzmq, pyyaml, pywinpty, python-json-logger, pygments, pycparser, psutil, prompt-toolkit, prometheus-client, platformdirs, parso, pandocfilters, packaging, overrides, nest-asyncio, mistune, MarkupSafe, jupyterlab-pygments, jsonpointer, json5, idna, h11, fqdn, executing, defusedxml, decorator, debugpy, colorama, charset-normalizer, certifi, babel, attrs, async-lru, terminado, rfc3339-validator, requests, referencing, python-dateutil, matplotlib-inline, jupyter-core, Jinja2, jedi, httpcore, comm, cffi, bleach, beautifulsoup4, asttokens, anyio, stack-data, jupyter-server-terminals, jupyter-client, jsonschema-specifications, httpx, arrow, argon2-cffi-bindings, jsonschema, isoduration, ipython, argon2-cffi, nbformat, ipykernel, nbclient, jupyter-events, nbconvert, jupyter-server, notebook-shim, jupyterlab-server, jupyter-lsp, jupyterlab, notebook
Successfully installed MarkupSafe-2.1.5 anyio-4.6.0 argon2-cffi-23.1.0 argon2-cffi-bindings-21.2.0 arrow-1.3.0 asttokens-2.4.1 async-lru-2.0.4 attrs-24.2.0 babel-2.16.0 beautifulsoup4-4.12.3 bleach-6.1.0 certifi-2024.8.30 cffi-1.17.1 charse
t-normalizer-3.3.2 colorama-0.4.6 comm-0.2.2 debugpy-1.8.6 decorator-5.1.1 defusedxml-0.7.1 executing-2.1.0 fastjsonsche
ma-2.20.0 fqdn-1.5.1 h11-0.14.0 httpcore-1.0.5 httpx-0.27.2 idna-3.10 ipykernel-6.29.5 ipython-8.27.0 isoduration-20.11.
0 jedi-0.19.1 Jinja2-3.1.4 json5-0.9.25 jsonpointer-3.0.0 jsonschema-4.23.0 jsonschema-specifications-2023.12.1 jupyter-
client-8.6.3 jupyter-core-5.7.2 jupyter-events-0.10.0 jupyter-lsp-2.25.0 jupyter-server-2.14.2 jupyter-server-terminals-0
.5.3 jupyterlab-4.2.5 jupyterlab-pygments-0.3.0 jupyterlab-server-2.27.3 matplotlib-inline-0.1.7 mistune-3.0.2 nbclient-
0.10.0 nbconvert-7.16.4 nbformat-5.10.4 nest-asyncio-1.6.0 notebook-7.2.2 notebook-shim-0.2.4 overrides-7.7.0 packaging-
24.1 pandocfilters-1.5.1 parso-0.8.4 platformdirs-4.3.6 prometheus-client-0.21.0 prompt-toolkit-3.0.48 psutil-6.0.0 pure
-eval-0.2.3 pycparser-2.22 pygments-2.18.0 python-dateutil-2.9.0.post0 python-json-logger-2.0.7 pywin32-306 pywinpty-2.0
.13 pyyaml-6.0.2 pyzmq-26.2.0 referencing-0.35.1 requests-2.32.3 rfc3339-validator-0.1.4 rfc3986-validator-0.1.1 rpds-py
-0.20.0 send2trash-1.8.3 setuptools-75.1.0 six-1.16.0 sniffio-1.3.1 soupsieve-2.6 stack-data-0.6.3 terminado-0.18.1 tiny
css2-1.3.0 tornado-6.4.1 traitlets-5.14.3 types-python-dateutil-2.9.0.20240906 uri-template-1.3.0 urllib3-2.2.3 wcwidth-
0.2.13 webcolors-24.8.0 webencodings-0.5.1 websocket-client-1.8.0
(.venv) C:\lab7>
```

3. Now try to check installed packages again. You should see more packages on your environment

```
(.venv) > pip list
```

```
(.venv) C:\lab7>pip list
Package Version
-----
anyio      4.6.0
argon2-cffi 23.1.0
argon2-cffi-bindings 21.2.0
arrow      1.3.0
asttokens  2.4.1
```

...

```
types-python-dateutil 2.9.0.20240906
uri-template          1.3.0
urllib3               2.2.3
wcwidth               0.2.13
webcolors              24.8.0
webencodings           0.5.1
websocket-client       1.8.0
(.venv) C:\lab7>
```

4. Now execute the following command to start the jupyter notebook. A new server will be initiated.

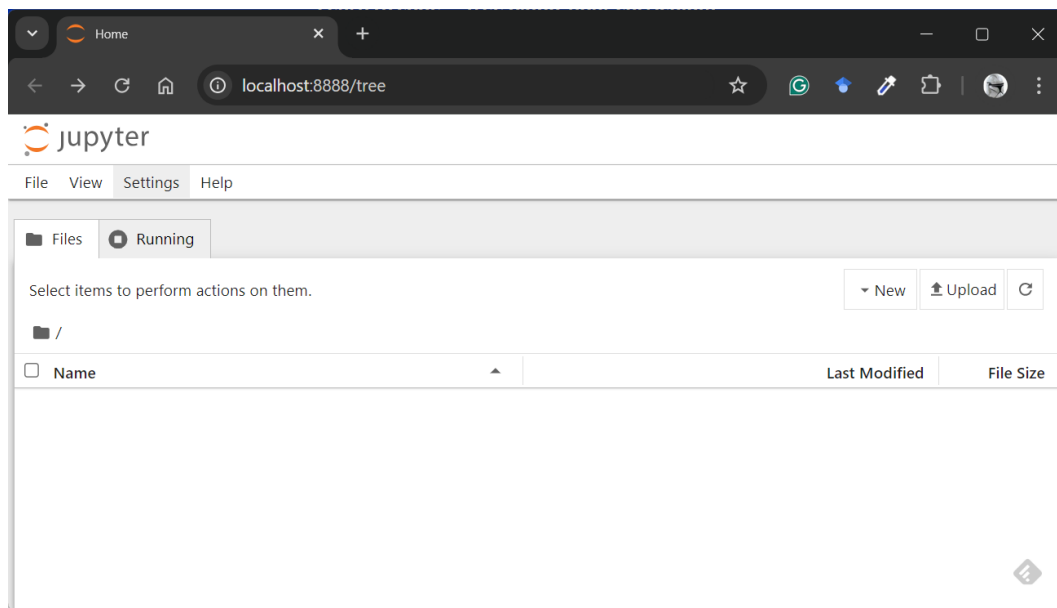
```
(.venv) > jupyter notebook
```

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```
(.venv) C:\lab7>jupyter notebook
2024-09-26 15:35:02.770 ServerApp] Extension package jupyter_lsp took 0.1305s to import
2024-09-26 15:35:02.906 ServerApp] jupyter_lsp | extension was successfully linked.
2024-09-26 15:35:02.917 ServerApp] jupyter_server_terminals | extension was successfully linked.
2024-09-26 15:35:02.928 ServerApp] jupyterlab | extension was successfully linked.
2024-09-26 15:35:02.939 ServerApp] notebook | extension was successfully linked.
2024-09-26 15:35:04.877 ServerApp] notebook_shim | extension was successfully linked.
2024-09-26 15:35:05.085 ServerApp] notebook_shim | extension was successfully loaded.
2024-09-26 15:35:05.095 ServerApp] jupyter_lsp | extension was successfully loaded.
2024-09-26 15:35:05.095 ServerApp] jupyter_server_terminals | extension was successfully loaded.
2024-09-26 15:35:05.099 LabApp] JupyterLab extension loaded from C:\lab7\venv\Lib\site-packages\jupyterlab
2024-09-26 15:35:05.099 LabApp] JupyterLab application directory is C:\lab7\venv\share\jupyterlab
2024-09-26 15:35:05.111 LabApp] Extension Manager is 'ppm'.
2024-09-26 15:35:05.276 ServerApp] jupyterlab | extension was successfully loaded.
2024-09-26 15:35:05.283 ServerApp] notebook | extension was successfully loaded.
2024-09-26 15:35:05.292 ServerApp] Serving notebooks from local directory: C:\lab7
2024-09-26 15:35:05.292 ServerApp] Jupyter Server 2.14.2 is running at:
2024-09-26 15:35:05.292 ServerApp] http://localhost:8888/tree?token=f3569e0c8f6c49391de7b7f7964ce225026ed4a84b4bcd0be
2024-09-26 15:35:05.292 ServerApp] http://127.0.0.1:8888/tree?token=f3569e0c8f6c49391de7b7f7964ce225026ed4a84b4bcd0be
2024-09-26 15:35:05.292 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).

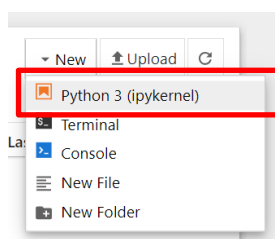
To access the server, open this file in a browser:
file:///C:/Users/chinann.ong/AppData/Roaming/jupyter/runtime/jupyter-42556-open.html
Or copy and paste one of these URLs:
http://localhost:8888/tree?token=f3569e0c8f6c49391de7b7f7964ce225026ed4a84b4bcd0be
http://127.0.0.1:8888/tree?token=f3569e0c8f6c49391de7b7f7964ce225026ed4a84b4bcd0be
2024-09-26 15:35:05.305 ServerApp] Skipped non-installed server(s): bash-language-server, dockerfile-language-server-nodejs, javascript-typescript-langserver, jedi-language-server, julia-language-server, py
right, python-language-server, python-lsp-server, x-language-server, sql-language-server, texlab, typescript-language-server, unified-language-server, vscode-css-language-server-bin, vscode-html-language-server-bin,
vscode-json-language-server-bin, yamll-language-server
```

a new tab on your browser will be open.

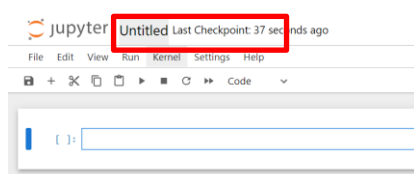


4. Create a simple notebook with markdown language

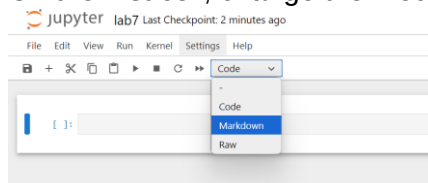
1. Create a new ipython notebook (.ipynb) by selecting Python 3 (ipykernel)



2. Rename the notebook as lab7.ipynb



3. On the first cell, change the mode to Markdown



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Then past the following code into it and press “Ctrl + Enter”

```
# Week 9: Exercise 7 - Virtual Environment and Jupyter Notebook  
*YOUR NAME*
```

```
## 1A2B (Bulls & Cows) Game Design/Development
```

4. Copy your code from last week exercise (1A2B (Bulls & Cows)) and paste into new cell. Where possible, include your code into separate cell by its functions and include the comments as the markdown on the top of each function definition.
5. Try to explore and test out other [markdown syntax](#) or [latex/scientific syntax](#) wrap with `$$ some latex syntax $$`. For example `$$ \frac{1}{2} $$` which will generate $\frac{1}{2}$. The Markdown cell in jupyter notebook works with [HTML syntax](#) too.

A sample.ipynb is provided with can be found at NTULearn

6. Clean up or cloning an environment (Optional)

1. To stop the jupyter notebook, you can either close the command prompt / terminal directly or press “ctrl + c” on the command prompt.
2. If you no longer need the entire virtual environment, you can delete the entire project folder i.e. lab7 and all the packages install will be removed.
3. If you wish to keep the package with its version or to clone another environment, you can create a “requirements.txt” file by executing the following command:

```
pip freeze > requirements.txt
```

You can then copy this file to another project folder with new environment and install all the packages with its version by executing the following command:

```
pip install -r requirements.txt
```

This command will install all packages listed in the requirement.txt file to your new environment.